

Root Cause Analysis

Sales Forecasting & Revenue Trend Analysis

Finding 1

The existing company forecast is biased low overall (−3.43% across 24 months), meaning actual revenue consistently beats the plan.

- Why 1: Why does actual revenue keep beating the forecast? Because festive-season months (Oct, Nov, Dec) show the largest positive variances: +11.1%, +10.3% and +2.4% in 2024, and +15.6%, +8.5% and +4.2% in 2025.
- Why 2: Why is the festive season under-forecast specifically? Because the seasonal index for Oct/Nov/Dec (1.49, 1.44, 1.44) is far above the average month (1.0), but the forecast appears to apply a flatter, less seasonally aggressive uplift.
- Why 3: Why would the forecast under-weight seasonality? Because forecasts are likely built from a base run-rate plus a standard percentage uplift, rather than from month-specific historical demand patterns.
- Why 4: Why hasn't this been corrected in two years of data? Because there is no visible feedback loop: no monthly accuracy or bias tracking exists to surface the pattern.
- Why 5 (root cause): Why is there no feedback loop? Because forecasting is treated as an annual planning exercise rather than a monitored, iterative process.

Root causes: [supported by data] Festive season is systematically under-forecast. [hypothesis to validate] Forecast-setting uses a flat seasonal uplift rather than month-level historical averages. [supported by data] No accuracy/bias tracking exists across the 24-month history.

Finding 2

Middle East is the only region where the forecast is too optimistic (−7.4% variance) and also the slowest-growing region (+11.2% YoY vs. +28.7% and +43.6% elsewhere).

- Why 1: Why is Middle East forecast too high? Because targets appear to assume the same growth trajectory as South Asia and Southeast Asia, but Middle East actual revenue is not keeping pace.
- Why 2: Why is Middle East growing more slowly? Because Middle East profit margin (32.0%) is comparable to the other two regions, so this isn't a margin or pricing problem; it points to demand or market share, not profitability.
- Why 3: Why would demand be flatter there specifically? Cannot be answered from this dataset. [hypothesis to validate] Needs market-level information: competitor activity, distribution reach, or category saturation in Saudi Arabia and UAE.

Root causes: [supported by data] Middle East growth is real and consistently slower, not a one-off dip. [hypothesis to validate] The forecast has not been adjusted region-by-region to reflect this gap.

Finding 3

A naive statistical forecasting model underperforms the company's existing forecast on backtest (69–79% accuracy vs. 94.8% for the company forecast, tested on the same 2025 holdout period).

- Why 1: Why does the statistical model do worse than the existing process? Because the model is trained on only 12 months of history (2024), too little to reliably estimate a trend or seasonal pattern.
- Why 2: Why is 12 months insufficient? Because a single festive season in the training data gets treated as 'the trend,' so the model over- or under-extrapolates it into the next year.
- Why 3: Why does the existing company forecast do better despite no visible statistical method? Because it appears to incorporate real business judgment (sales rep and regional manager input) that a 12-point regression can't capture.

Root causes: [supported by data] Data history is too short for a standalone statistical model to add value yet. [hypothesis to validate] Human judgment in the current process is capturing information (market conditions, deal pipeline) not present in the dataset.

Root Cause Summary

Branch	Root Cause	Supported by Data?
People	Forecast-setting likely relies on manager judgment with no structured region/season adjustment	Partially (inferred)
Process	No monthly forecast-vs-actual review or feedback loop	Yes
Data	Only 24 months of history limits statistical modeling	Yes
Method	Seasonal uplift, where used, appears flatter than actual seasonal swings	Hypothesis to validate
Market	Middle East demand growth is structurally slower	Hypothesis to validate